

Madhulika Vikraman

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SUMMARY

MS Computer Science student with professional experience building Python, SQL, and Go software systems, data pipelines, APIs, and automation tools. Experienced applying machine learning, NLP, LLMs, and retrieval techniques to real-world data analysis and modeling problems using PyTorch, TensorFlow, scikit-learn, and related frameworks.

EDUCATION

University of Southern California

M.S. in Computer Science, GPA 3.8/4.0; Grader, CSCI 585 Database Systems; GRIDS Club

Los Angeles, CA

Aug. 2025 - May 2027

PES University, RR Campus

B.Tech in Computer Science, GPA 9.29/10; 4-semester merit scholarship recipient

Bangalore, India

Dec. 2020 - July 2024

WORK EXPERIENCE

Student Research Engineer, USC Norman Lear Center

May 2026 - Present

Ongoing

Los Angeles, CA

- Built a Python NLP pipeline around locally served LLMs (Ollama) that automated qualitative coding, statistical analysis, and inter-rater reliability computation, producing research-grade data visualizations.
- Applied LLM-based sentiment analysis with a human-validated research codebook, adapting and evaluating deep learning models for structured analysis tasks.

Associate Software Engineer, Celestica

Sept. 2024 - July 2025

SONiC

Chennai, India

- Designed and implemented production CLI features for SONiC in Python, Go, YANG, and Click/Klish, translating declarative data models into generated C and Python interfaces.
- Debugged and resolved critical, cross-layer bugs spanning CLI, data-model, and backend code paths across multiple releases, reducing release-time issues by 30%.
- Built Ansible-based automation to generate and execute state-changing device configurations, improving operational efficiency, scalability, and stability by 50% and eliminating manual intervention in a production toolchain.

Software Development Intern, Sandvine Technologies (Applogic Networks)

Jan. 2024 - Aug. 2024

Bangalore, India

- Rebuilt a data conversion/transport feature in Python with concurrent data collection and analysis, profiling and optimizing throughput by over 60%.
- Developed Python, SQL, and REST API modules for an RDBMS-backed network traffic classification platform, supporting transformation and analysis of production network data.
- Built a client-server system in Go to simulate real-time network traffic using custom binary structures parsed from Parquet files, streamed over raw TCP sockets.

ACADEMIC & RESEARCH PROJECTS

Pavilion Cerebro, Novi-Infra | *Multi-Agent AI, Swarm* |

4-day startup sprint hackathon(Tiehub), team of 6

- Built a multi-agent building health monitor on BMS telemetry and sensor data: a swarm of four specialized agents (crack detection via CV, concrete health from UPV data, NASA Li-ion battery aging, ASHRAE HVAC faults) emit 0 to 1 risk scores that a coordinator ranks by risk, consequence, and time, replayed live on a dashboard.
- Reached 92.9% crack-detection accuracy on 1,600 held-out images and 96.9% HVAC diagnosis accuracy with 1.5% false alarms; interviewed 11 engineers and architects to define 11 problem areas and the \$599/month per-building model.

Comparison of Biologically Inspired Neural Architectures | *NLP, LLMs, Research*

Ongoing

- Conducting a controlled comparison of biologically-inspired architectures (Hopfield attention, spiking neural networks) versus Transformer baselines on NLP tasks, evaluating accuracy, robustness, and efficiency tradeoffs.

Legal LLM, USC GRIDS | *FAISS, RAG, LLMs*

Ongoing

- Building a RAG pipeline (FAISS plus multiple LLMs) to identify risky clauses in lease agreements, including context-window and retrieval-latency tradeoff analysis.

Liquid Neural Networks for Heart Disease & Seizure Prediction | *PyTorch, Time-Series*

Research publication pending

- Implemented and benchmarked Liquid Time-Constant Networks and Neural ODEs in PyTorch against standard deep learning baselines for multivariate EEG/EKG time-series prediction for a research paper.

Sarcasm, Irony, and Humor Detection in Low-Resource Languages | *PyTorch, TensorFlow, Keras*

Capstone, team of 4

- Trained and evaluated 30 ML/DL models for low-resource text classification, achieving 86 to 89% accuracy; presented at ICICV 2024 and published in Springer LNNS.

TECHNICAL SKILLS

Languages, Frameworks: Python, C++, C, Go, Bash, YANG, Java, SQL, MySQL, R, React, JavaScript, Spring, Redis, MongoDB

AI/ML Frameworks: PyTorch, TensorFlow, Keras, HuggingFace, Scikit-learn, LLMs, LangChain, LangGraph, RAG

Toolchain / Systems: CLI/code-generation, Click/Klish, Ansible, Linux, REST APIs, TCP/socket-level client-server systems, Git, Jira (Agile), Docker, Kubernetes, Jenkins, Azure

Coursework: Analysis of Algorithms, Deep Learning and its Applications, Machine Learning, Operating Systems, Computer Networks, Databases, Data Structures, Statistics, Reinforcement Learning